

# EXPERIMENT 36

## Digital Indicator Extraction

### AIM

To extract unique URLs, email addresses, IPv4 addresses, and domain names from a recovered text file.

### ALGORITHM

1. Read the text file.
2. Use regular expressions for URLs, emails, IPv4 addresses, and domains.
3. Remove duplicate values.
4. Store each category separately.
5. Display the values and counts.

### CODE

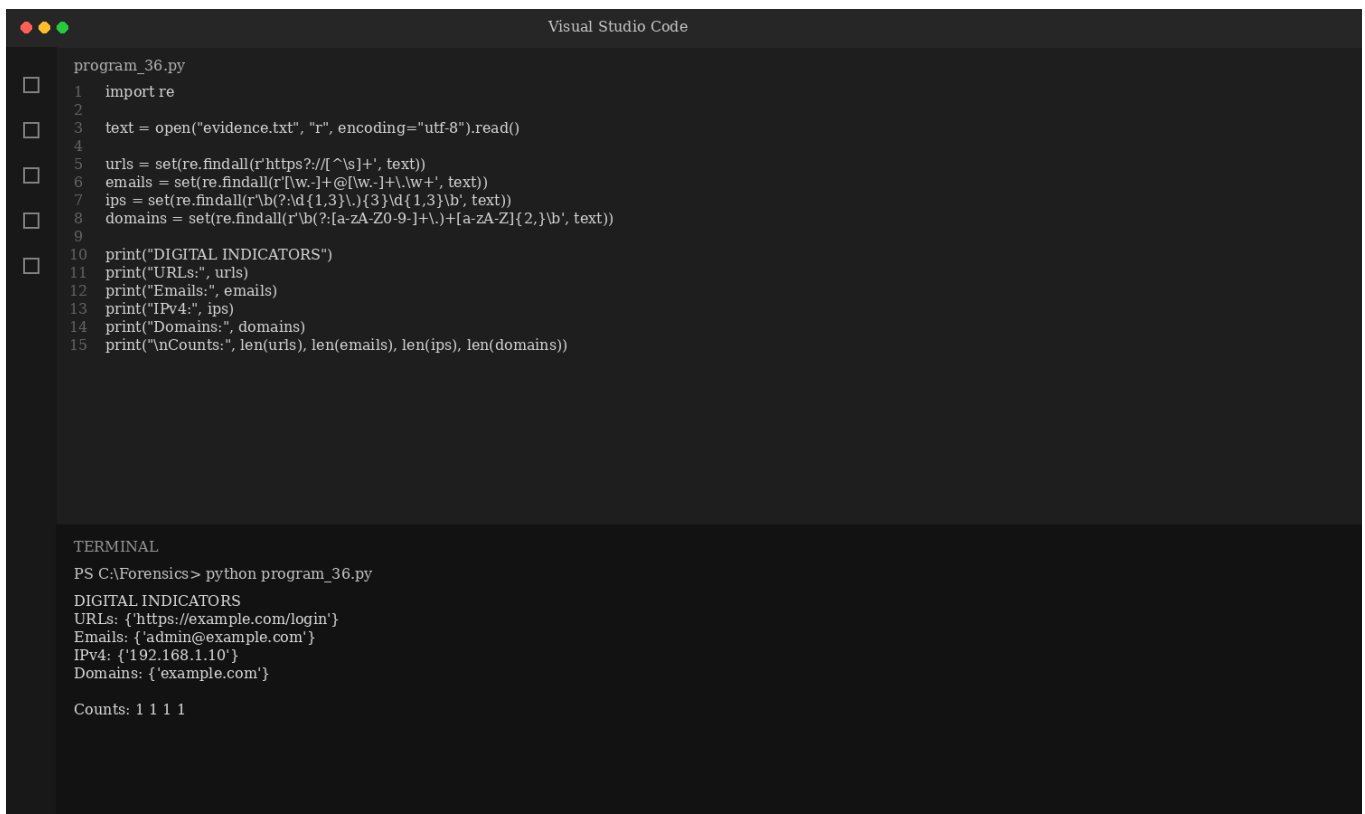
```
import re

text = open("evidence.txt", "r", encoding="utf-8").read()

urls = set(re.findall(r'https?://[^\s]+', text))
emails = set(re.findall(r'[w.-]+@[w.-]+\w+', text))
ips = set(re.findall(r'\b(?:\d{1,3}\.){3}\d{1,3}\b', text))
domains = set(re.findall(r'\b(?:[a-zA-Z0-9-]+\.[a-zA-Z]{2,})\b', text))

print("DIGITAL INDICATORS")
print("URLs:", urls)
print("Emails:", emails)
print("IPv4:", ips)
print("Domains:", domains)
print("\nCounts:", len(urls), len(emails), len(ips), len(domains))
```

### OUTPUT



```
program_36.py
1  import re
2
3  text = open("evidence.txt", "r", encoding="utf-8").read()
4
5  urls = set(re.findall(r'https?://[^\s]+', text))
6  emails = set(re.findall(r'[w.-]+@[w.-]+\w+', text))
7  ips = set(re.findall(r'\b(?:\d{1,3}\.){3}\d{1,3}\b', text))
8  domains = set(re.findall(r'\b(?:[a-zA-Z0-9-]+\.[a-zA-Z]{2,})\b', text))
9
10 print("DIGITAL INDICATORS")
11 print("URLs:", urls)
12 print("Emails:", emails)
13 print("IPv4:", ips)
14 print("Domains:", domains)
15 print("\nCounts:", len(urls), len(emails), len(ips), len(domains))

TERMINAL
PS C:\Forensics> python program_36.py
DIGITAL INDICATORS
URLs: {'https://example.com/login'}
Emails: {'admin@example.com'}
IPv4: {'192.168.1.10'}
Domains: {'example.com'}

Counts: 1 1 1 1
```

### RESULT

The program extracts unique URLs, email addresses, IPv4 addresses and domains from recovered text.